

Technical Report

Predicting Football Match Outcomes Using Half-Time Statistics

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1 Dataset Description

1.1 Source and domain

The dataset consists of football match statistics from two consecutive seasons (2023–2024 and 2024–2025) corresponding to the Spanish first division league (LaLiga). Each instance represents a single match and contains quantitative performance statistics. The dataset belongs to the sports analytics domain and is publicly available.

The two seasons were combined to increase the number of samples and improve model generalization.

1.2 Target variable

The target variable is *FTR* (*Full Time Result*), a categorical variable with three classes: home win (H), draw (D), and away win (A). This defines a multiclass supervised classification problem.

1.3 Input features

The input features used in this study include match statistics available at half-time for both home and away teams. In addition, several derived features were computed as differences between home and away team statistics in order to capture relative performance.

Feature	Description	Type
HTHG	Home team goals at half-time	Original
HTAG	Away team goals at half-time	Original
HS	Home team shots	Original
AS	Away team shots	Original
HST	Home team shots on target	Original
AST	Away team shots on target	Original
HF	Home team fouls committed	Original
AF	Away team fouls committed	Original
HC	Home team corners	Original
AC	Away team corners	Original
HY	Home team yellow cards	Original
AY	Away team yellow cards	Original
HR	Home team red cards	Original
AR	Away team red cards	Original
GoalDiff_HT	Goal difference at half-time (HTHG – HTAG)	Derived
ShotDiff	Shot difference (HS – AS)	Derived
ShotOnTargetDiff	Shot on target difference (HST – AST)	Derived
FoulDiff	Foul difference (HF – AF)	Derived
CornerDiff	Corner difference (HC – AC)	Derived
CardDiff	Card difference (HY+HR – AY+AR)	Derived

Table 1: Original and derived input features used for model training.

These features constitute the final input space used for model training and evaluation.

1.4 Class distribution

Figure 1 shows the distribution of match outcomes. Home wins are the most frequent class, followed by away wins and draws, indicating a moderate class imbalance.

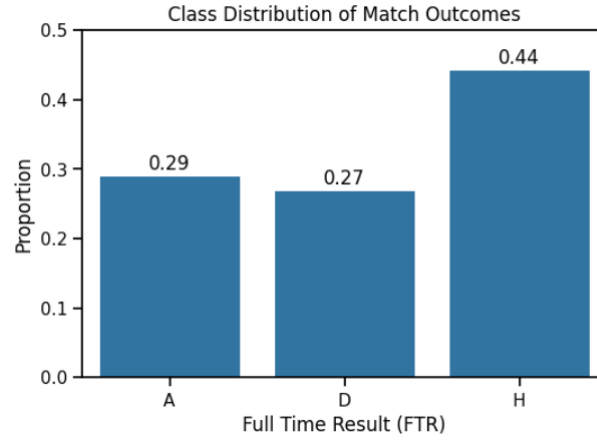


Figure 1: Class distribution of the target variable (FTR).

2 Machine Learning Pipeline

2.1 Data preprocessing

Only statistics available at half-time were used in order to avoid information leakage. No extensive preprocessing was applied, in line with the assignment guidelines. Additional features were engineered as differences between home and away team statistics (e.g., goal difference and shot difference).

2.2 Train-test split

The dataset was split into training and test sets using a 70/30 ratio with stratified sampling to preserve class distribution.

2.3 Models tested

Three classification models were evaluated: Logistic Regression, Random Forest, and Gradient Boosting. These models allow comparing a linear baseline with non-linear ensemble methods.

2.4 Hyperparameter tuning

No extensive hyperparameter tuning was performed. The focus of the study was model comparison rather than optimization. Hyperparameter tuning is considered future work.

3 Evaluation

3.1 Confusion matrices

Confusion matrices were computed for all models to analyze class-wise performance. Logistic Regression showed the most stable behavior across classes.

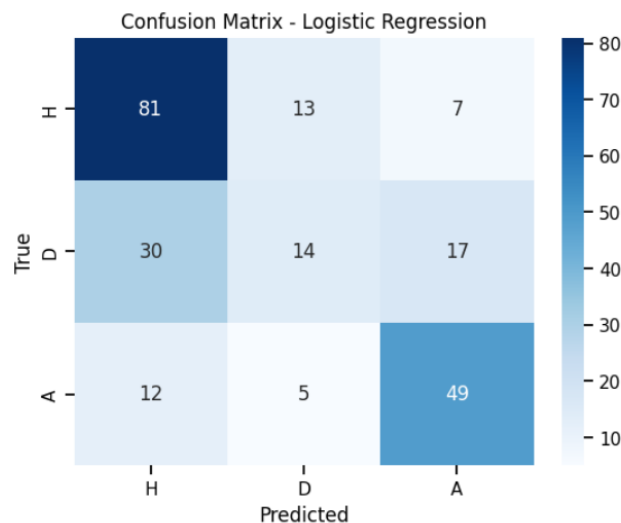


Figure 2: Confusion matrix for Logistic Regression.

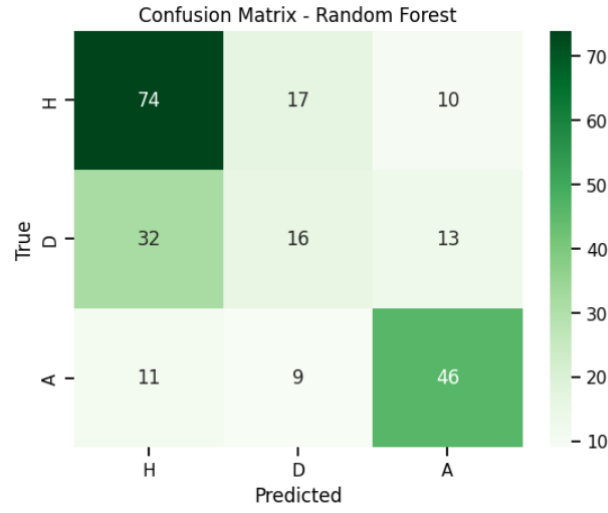


Figure 3: Confusion matrix for Random Forest.

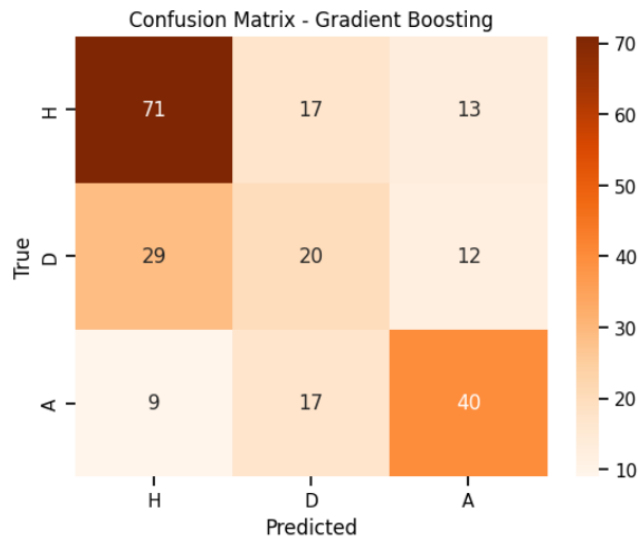


Figure 4: Confusion matrix for Gradient Boosting.

3.2 Precision, Recall and F1-score

Precision, recall and F1-score were computed for each class. Draws consistently showed lower recall across all models, reflecting their intrinsic uncertainty.

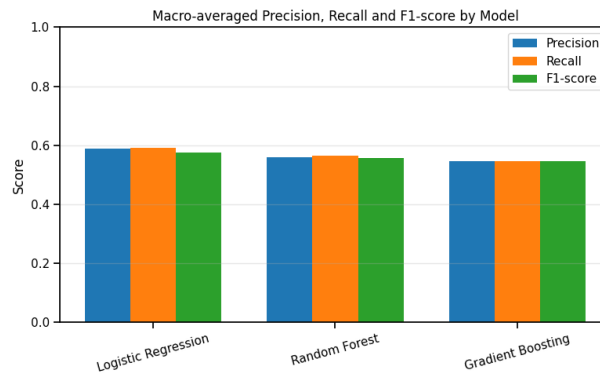


Figure 5: Macro-averaged Precision, Recall and F1-score for the evaluated models.

Figure 5 compares macro-averaged Precision, Recall and F1-score across models, confirming the superior overall balance achieved by Logistic Regression.

3.3 ROC curve and AUC

Receiver Operating Characteristic (ROC) curves were computed using a One-vs-Rest strategy. The next figure shows the ROC curves for the final selected model.

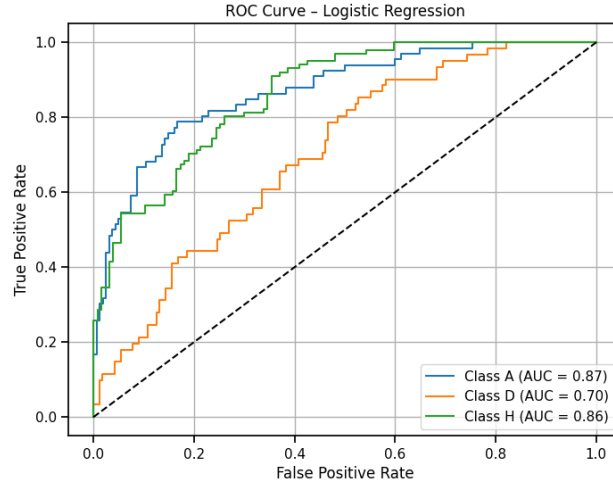


Figure 6: ROC curve for Logistic Regression

3.4 Precision-Recall curve

A Precision-Recall curve was computed for the final model to further evaluate performance under class imbalance.

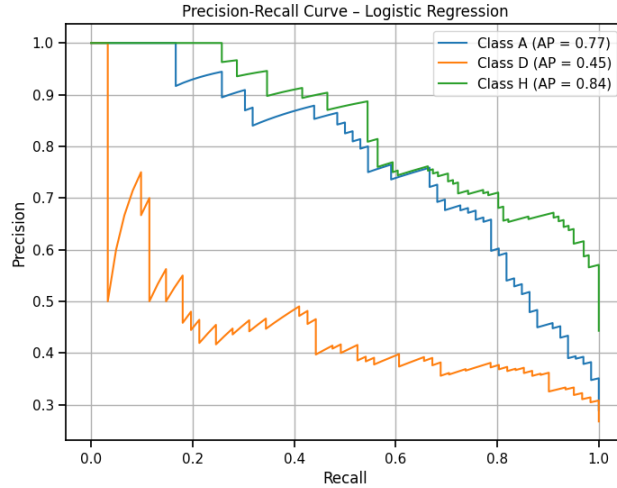


Figure 7: Precision-Recall curve for Logistic Regression.

3.5 Model comparison

Table 2 summarizes the macro-averaged ROC AUC obtained by each model.

Model	ROC AUC (macro)
Logistic Regression	0.812
Random Forest	0.769
Gradient Boosting	0.758

Table 2: Model comparison using macro-averaged ROC AUC.

4 Conclusions

Logistic Regression achieved the best overall performance in terms of macro-averaged ROC AUC while maintaining high interpretability. The results show that half-time statistics provide strong predictive signals for match outcomes. Goal difference and shot-related statistics at half-time were found to be the most informative predictors, highlighting the importance of early match dominance.

Draws remain the most difficult class to predict due to their inherent uncertainty. Future work could incorporate contextual pre-match information, perform hyperparameter tuning, and apply interpretability techniques such as SHAP.